Question:

Toulouse has twice as many sheep as Charleston. Charleston has 4 times as many sheep as Seattle. How many sheep do Toulouse, Charleston, and Seattle have together if Seattle has 20 sheep?

Reasoning + Answer:

<think>To solve this problem, let's first define a variable to help us work through the information given. Let's say \(S \) represents the number of sheep in Seattle. According to the problem, \(S = 20 \). So, Seattle has 20 sheep. Next, we need to find the number of sheep in Charleston. The problem states that Charleston has 4 times as many sheep as Seattle. Since we know Seattle has 20 sheep, we can express the number of sheep in Charleston using the variable \(C \). So, we can write the equation for Charleston as \(C = 4 \times S \). Substituting the value of \(S \) (which is 20) into the equation gives us \(C = 4 \times 20 \). Now, let's calculate \(4 \times 20 \): \(4 \times 20 = 80 \). Therefore, Charleston has 80 sheep. Now, let's focus on Toulouse. The problem states that Toulouse has twice as many sheep as Charleston. Let's denote the number of sheep in Toulouse as \(T \). According to the given information, \(T = 2 \times C \). Since we determined that Charleston has 80 sheep (\(C = 80 \)), we can substitute this value into the equation to find \(T \): \(T = 2 \times 80 \). Now, let's calculate \(2 \times 80 \): \(2 \times 80 = 160 \). So, Toulouse has 160 sheep. Finally, to find the total number of sheep in Toulouse, Charleston, and Seattle together, we need to add the number of sheep from all three cities. We already know \(S = 20 \) for Seattle, \(C = 80 \) for Charleston, and \(T = 160 \) for Toulouse. So, we add these numbers together: \(20 + 80 + 160 \). Let's perform this addition: \(20 + 80 = 100 \), and then \(100 + 160 = 260 \).</think><answer> 260</answer>